

SIH 2026 — Day 1 Task 1: Master Summary Document

Problem Statement ID: PS 26117 / SIH26117 **Title:** Sovereign On-Premise Agentic AI Workbench using Open-Weight Multimodal LLMs for Confidential Industrial Work **Organization:** Mangalore Refinery and Petrochemicals Limited (MRPL) **Category:** Software | **Theme:** Smart Automation

Executive Summary

This documentation repository contains the complete **Day 1 Task 1: Problem Understanding, Solution Planning & System Design** package for SIH 2026.

The **Sovereign AI Workbench** is a 100% local, air-gapped, privacy-first AI platform engineered specifically for heavy industrial confidential environments like **MRPL refineries**. It eliminates external cloud telemetry and data leakage by running quantized open-weight LLMs (**qwen3.5:4b**, **qwen2.5-coder:7b**, **glm-ocr:q8_0**, **nomic-embed-text**) entirely on local edge hardware via Ollama and FastAPI.

Day 1 Documentation Index

#	Document	Target Section / Focus Area
01	01_problem_statement_understanding.md	MRPL Context, Industrial Pain Points, Air-Gap Constraints & Scope
02	02_proposed_solution.md	Solution Core, Operational Capabilities & Anti-Hallucination Guardrails
03	03_unique_key_features.md	Key Innovations, Multimodal OCR Pipeline, In-Memory Cosine RAG Engine
04	04_target_users.md	Industrial User Personas (Plant Engineers, Safety Officers, IT, Executives)
05	05_system_architecture.md	High-Level Architecture Diagram, Layer Breakdown & Component Specs
06	06_workflow_user_flow_diagram.md	Interactive Sequence Diagrams (OCR Ingestion, RAG Grounding, Code Generation)
07	07_data_flow_diagram.md	DFD Level 0 (Context Diagram), Level 1 (System Processes), Level 2 (OCR/RAG)
08	08_module_definition.md	Software Modules (UI Gateway, FastAPI Router, Vector Engine, Model Runtime)
09	09_technology_stack.md	Open-Weight Models Matrix, Tech Justification & Hardware Specs

Mentor Review Alignment Matrix

Mentor Evaluation Metric	Addressed In Document	Key Highlights
Problem–Solution Alignment	01_problem_statement_understanding.md, 02_proposed_solution.md	Tailored specifically for MRPL refinery SOPs, plant safety, and confidential blueprints.

Technical Feasibility	09_technology_stack.md, 05_system_architecture.md	100% working local stack with Ollama, FastAPI, and quantized models on consumer hardware.
System Architecture	05_system_architecture.md	Modular 4-tier design: Frontend, FastAPI Gateway, Cosine Vector Engine, Local Model Runtime.
Workflow / User Flow	06_workflow_user_flow_diagram.md	Clear flowcharts and sequence diagrams for Document Upload, RAG Querying, and OCR processing.
Data Flow Diagram (DFD)	07_data_flow_diagram.md	Complete DFD Level 0, Level 1, and Level 2 pipeline mappings.
Scalability & Security	03_unique_key_features.md, 09_technology_stack.md	Air-gapped network isolation with zero telemetry; scalable through modular tool definitions.
Module Clarity	08_module_definition.md	Clean separation of concerns between web frontend, backend routing, OCR, RAG, and LLMs.
Innovation & Uniqueness	03_unique_key_features.md	Local GLM-OCR multimodal pipeline + lightweight in-memory cosine vector store with zero DB overhead.

■ Quick Verification & Demo Status

- ■ **Local Model Execution:** qwen3.5:4b (General/RAG), qwen2.5-coder:7b (Code), glm-ocr:q8_0 (OCR), nomic-embed-text (Embeddings) running locally via Ollama.
- ■ **Local RAG Pipeline:** Grounded vector search over uploaded PDFs and text files with strict fallback when info is missing.
- ■ **Local OCR Ingestion:** Scanned image text extraction via GLM-OCR feeding into the same unified vector knowledge index.
- ■ **FastAPI Backend + Web UI:** Functional web interface served directly from local Python backend.